Does Functional Similarity Induced by Distillation Preserve Internal Representations and Explanatory Mechanisms in Autoregressive Models?

1st Martin Luca Perciante
dept. name of organization (of Aff.)
Universidad Católica del Uruguay
Montevideo, Uruguay
martinluca.perciante@ucu.edu.uy

2nd Given Name Surname
dept. name of organization (of Aff.)
Universidad Católica del Uruguay
Montevideo, Uruguay
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
Universidad Católica del Uruguay
Montevideo, Uruguay
email address or ORCID

***Abstract*—Knowledge distillation [4] is often evaluated in terms of functional similarity between teacher and student models, typically measured through predictive performance. However, it remains unclear whether this functional similarity also implies the preservation of internal representations and explanatory mechanisms without explicitly imposing them. This work investigates this question in autoregressive Transformers. A TinyLlama [16] teacher is distilled into a family of 11 attention layer students with different training configurations. Internal representations are compared through layer-wise attention-map metrics (cosine similarity, Frobenius distance, Jensen–Shannon divergence [6], spectral effective rank [9], [11]), per-head similarity, while explanatory mechanisms are evaluated through LIME [10], SHAP [7], and Integrated Gradients [13] (zero-vector baseline), moving beyond qualitative examples to aggregate agreement (Jaccard overlap, rank correlation, sign agreement) and deletion-based faithfulness metrics [2]. The study is conducted on binary IMDb [8] sentiment classification and on four-class AG News [17] topic classification. Results show that distillation does not transfer the teacher's explanatory mechanisms on either dataset, even though it robustly and monotonically improves the student's distributional fidelity to the teacher on both datasets. Representational similarity, in turn, is only partially and inconsistently attributable to distillation: attention-level and per-head metrics show that a non-distilled baseline is as close to the teacher as the distilled students. These findings suggest that the observed similarity to the teacher is driven more by the shared data and task than by distillation itself, indicating that functional similarity does not, by itself, guarantee the preservation of a teacher's explanatory mechanisms or internal representations.**

***Index Terms*—Distillation, Attention, XAI, Explainability**

I. Introduction

Explainable Artificial Intelligence (XAI) seeks to interpret and explain the behavior of complex models, essential in domains where deployed systems must justify autonomous decisions and foster user trust. This task is particularly difficult for Large Language Models (LLMs), where probabilistic text generation replaces classification with a single class label output.

Autoregressive Transformer-based models are the architecture of choice for this kind of modeling in current Natural Language Processing (NLP), driving recent progress largely by growing in parameters and complexity; the resulting computational cost has motivated compression techniques such as knowledge distillation (KD), in which a smaller model (student) is trained to approximate the output distribution of a larger model (teacher) [4], alongside dataset training.

Standard KD is centered on functional similarity: the same input is made to produce a similar output, regardless of the internal computations that produce it. Prior work has shown that different attention configurations can induce similar functional behavior [5], raising the question of whether KD enforces a preference for the teacher's internal representations, or whether it can converge to different attention maps while reproducing the teacher's output distribution. Bringing XAI into this picture, different models can provide different explanations for the same example while achieving similar performance over a dataset, so this question extends naturally to explanatory mechanisms as well.

This work investigates whether the functional similarity induced by logit-based KD extends to similarity in internal representations and explanatory mechanisms. We adopt a generative classification setting, in which target classes correspond to single vocabulary tokens, allowing internal representations to be compared directly between teacher and student. We conducted the experiments on binary **IMDb** sentiment classification and in **AG News** four-class topic classification.

This work contributions are:

- We show that logit-based KD, even when it reproduces the teacher's output distribution with increasing fidelity, does not correspondingly transfer its explanatory mechanisms.
- We show that representational similarity to the teacher (attention and per-head geometry) is not a reliable signature of distillation itself.
- We show this dissociation holds across both a binary (**IMDb**) and a four-class (**AG News**) classification task, indicating it is not an artifact of a single dataset or label space.

This research was funded by the Agencia Nacional de Investigación e Innovación (ANII), Uruguay, under code POS_NAC_2025_4_189264.

II. RELATED WORK

Knowledge distillation (KD) transfers information from a high-capacity teacher to a smaller student by adding a Kullback–Leibler (KL) divergence term to the loss function, often softened by a temperature T [4]. KD is widely used in deep networks and large language models, but its success is evaluated almost exclusively via functional metrics such as accuracy [3]. Some works explicitly bias the loss towards representational alignment, such as Attention Distillation [18], which directly supervises student attention maps against the teacher's.

Post-hoc explainability methods such as **LIME** [10] and **SHAP** [7] treat the model as a black box, while gradient-based methods such as **Integrated Gradients** [13] exploit differentiability; all three are commonly used to quantify token-level contributions to a prediction. Beyond post-hoc attribution, attention maps offer a potentially interpretable internal mechanism, but whether they constitute a faithful explanation remains contested: Jain and Wallace [5] show that alternative attention configurations can induce similar predictions, whereas Wiegreffe and Pinter [15] argue attention can still provide plausible explanations under appropriate conditions.

III. EXPERIMENTS

A. Experimental Setup

The project used **Python**, **PyTorch**, Hugging Face's **transformers**, **lime/shap** for explanation, **SciPy** for the rank/value-correlation statistics used in the aggregate explanation metrics, and **scikit-learn** for evaluation metrics.

The teacher is **TinyLlama-1.1B-Chat-v1.0** [16], a 22-layer decoder-only Transformer [14]. Every student shares the teacher's embeddings, attention mechanism, and hidden dimensions, reducing only depth to 11 randomly-initialized blocks (2:1 Transformer layer correspondence; compression rate $CR \approx 1.787$, $\approx 44\%$).

The studied problem is reformulated as a generative classification [12], where class-labels are single vocabulary tokens. Dataset inputs are inserted into a fixed prompt template, and only the class-token logits are compared at inference. For **IMDb** [8] (50000 reviews, labels *positive*/*negative*), the template is `"Review: {input} Summary:"`; filtered to the teacher's 2048-token context, splits are 22492/2500/24995 (train/val/test). For **AG News** [17] (4 classes: *World*, *Sports*, *Business*, *Sci/Tech*), the template is `"Article: {input} Theme:"`. Since *Sci/Tech* tokenizes to four sub-tokens, it is relabeled *Science* (one token) purely to preserve the single-token framing. Splits are 30000/100/7600 (balanced). All other hyperparameters are shared across datasets.

The teacher is fully fine-tuned per dataset for one epoch (lr 2×10^{-5}, batch size 1, AdamW, early stopping after 5 non-improving validation rounds). Every student is randomly initialized and trained for 3 epochs (lr 1×10^{-4}, no early stopping), minimizing (1).

$$\mathcal{L} = \alpha\,\mathcal{L}_{CE} + (1-\alpha)\,T^2 D_{KL}\left(P_P \| P_S\right) + \beta\,T^2 D_{KL}\left(P_P^{(c)} \| P_S^{(c)}\right) \quad (1)$$

with $T = 2$ fixed, $P_P^{(c)}, P_S^{(c)}$ the probability distributions restricted to the class-label tokens, and (α, β) set per Table I, defining a family of four students.

Explanations from **LIME**, **SHAP**, and **Integrated Gradients** (zero-embedding baseline) are compared between teacher and student via aggregate metrics, computed over $N = 100$ held-out examples per dataset for all three methods. Representational comparisons use the 2:1 teacher-student layer correspondence: attention maps are averaged over held-out samples ($N = 200$) at each layer, and student layer i is then compared against the average of teacher layers $2i$ and $2i + 1$. All experiments ran on an NVIDIA RTX 5070 Ti (16GB).

B. Comparison Metrics

We compare Attention maps with complementary geometric, probabilistic, and spectral measures. Given two attention matrices A, B and their vectorizations $\vec{A}, \vec{B}$, we compared them via *cosine similarity* $\cos(\theta) = \langle \vec{A}, \vec{B} \rangle / (\|\vec{A}\|_2 \|\vec{B}\|_2) \in [-1, 1]$, the *Frobenius distance* $d_F(A, B) = \|A - B\|_F = \sum_{i,j} (A - B)_{ij}^2$ and the row-averaged *Jensen–Shannon divergence* $\frac{1}{n} \sum_i JS(A_i || B_i)$ were A_i is the i_{th} row of A. Since Attention maps are $n \times n$ row-stochastic matrices, the *Frobenius distance* is bounded by $\sqrt{2n}$ (64 for our $n = 2048$ maps). The *Jensen–Shannon divergence* is theoretically bounded in $[0, \ln 2]$ if measured in nats [6], where 0 means equal distributions. For spectral metrics we propose the entropy-based *effective rank* [11] and the *Participation Ratio* [9].

For evaluating explainability methods, we compute aggregate metrics over held-out samples, comparing each student's word-level attribution against the teacher's for every method. Metrics include top-k *Jaccard overlap* ($k = 10$), *Spearman correlation* of signed scores over the common attributed vocabulary, and deletion-based *faithfulness*@k [2], the drop in true-class probability after masking the top-k words (computed per model, including the teacher, as reference). Given **LIME's** stochastic nature, we also measure *stability* as the Jaccard@10 mean across 5 re-runs with different seeds.

C. Evaluation on the ***Test*** Set

Table II reports classification performance and average KL divergence to the teacher ($T = 1, 2$) for all models on both datasets. On **IMDb**, distillation's effect on accuracy is mixed:

Table I: Model family. All students share the same reduced (11-layer) architecture and differ only in (α, β).

Model	Init.	α	β	Description
P (teacher)	pretrained	–	–	fine-tuned, CE only
B	random	1	0	no distillation
S_{bad}	random	0.9	0	weak global KD
S_1	random	0.5	0	standard global KD
S_2	random	0.5	0.25	+ class-restricted KD

Table II: Accuracy / F1 (macro-averaged for **AG News**) on the **test** set, and average KL divergence to the teacher P at $T = 1, 2$ (**IMDb**: $N = 3000$ balanced subset; **AG News**: full **test** set).

Dataset	Model	Acc.	F1	$KL_{T=1}$	$KL_{T=2}$
IMDb	P	0.954	0.954	–	–
IMDb	B	0.858	0.857	2.847	1.686
IMDb	S_{bad}	0.813	0.810	0.104	0.132
IMDb	S_1	0.798	0.796	0.104	0.127
IMDb	S_2	0.867	0.867	0.111	0.193
AG News	P	0.849	0.848	–	–
AG News	B	0.886	0.886	2.080	1.784
AG News	S_{bad}	0.883	0.884	0.630	0.418
AG News	S_1	0.878	0.878	0.279	0.229
AG News	S_2	0.880	0.881	0.248	0.247

only the class-restricted S_2 edges out the non-distilled B (0.867 vs. 0.858), while S_{bad} and S_1 both *underperform* B (0.813 and 0.798), i.e. the distillation term hurts relative to training from scratch. On **AG News**, this pattern is more pronounced: B achieves the *highest* accuracy of all five models ($B > S_{bad} > S_2 > S_1 > P$). KL fidelity, by contrast, follows the same monotonic pattern on *both* datasets and temperatures: B is by far the least faithful, improving with distillation strength ($B > S_{bad} > S_1 \gtrsim S_2$). Notably, on **IMDb** S_1 attains *lower* KL than S_2 despite worse accuracy: global KL divergence is not a predictive proxy for classification performance, nor, as shown below, of explanatory or representational alignment.

D. Functional Evaluation and Explanation Comparison

LIME explanations for a representative **AG News** *Business* example show the teacher's top attributions concentrated on topical terms (*pension*, *workers*, *Unions*, *Federal*); students share most of these words, mixing in generic phrase tokens absent from the teacher's list. A similar pattern holds on **IMDb**, where the teacher and most students concentrate on genuine sentiment words (*extraordinary*, *wonderful*, *dull*, *clichéd*), except S_1, which consistently favors function words (*it*, *and*, *the*, *to*) over content words regardless of its classification performance.

SHAP explanations for the same examples are far less concentrated: the top-10 words capture only a small fraction of the total attribution, with the remaining $\sim$140 tokens accounting for most of it, and content/function words appear interspersed across all five models rather than following a clean teacher/student split. **Integrated Gradients**, for every model including the teacher, is instead dominated by one or two extreme-magnitude spikes on semantically arbitrary tokens (proper nouns, subword fragments) rather than topical or sentiment content.

Table III reports **LIME** stability (mean pairwise Jaccard@10 across 5 re-runs, $N = 20$). The teacher is more stable than the students on **AG News** but not always on **IMDb**, so stability alone is not a proxy for explanation quality.

Table IV reports aggregate **LIME** agreement (vs. teacher) and faithfulness on both datasets, $N = 100$, $k = 10$. On both, S_2 is consistently the best-aligned and most-faithful student, though the gap to B is modest.

SHAP (Table IV) shows the same S_2-best ranking, but faithfulness near zero for *every* model, including the teacher – likely a limitation of masking-based perturbation rather than a genuine absence of faithful explanations.

Integrated Gradients (Table IV) shows high top-10 token overlap ($\approx$ 0.78–0.89) alongside near-zero or negative rank correlations, indicating agreement on *which* tokens matter without agreement on their relative importance. Sign agreement@10 (not tabulated) mirrors this split: **LIME/SHAP** follow the same S_2-best pattern (0.72–0.82), whereas IG sits at chance level (0.50–0.54) for every model, so its cross-model consistency is comparatively shallow.

E. Representational Comparison

Table V reports cosine similarity, Frobenius distance, and Jensen–Shannon divergence between each student's attention map and the teacher's average (P_{avg}, matched via the 2:1 correspondence), across six layers per dataset (full 11-layer tables are in the extended version). On **IMDb**, B is slightly lower than the three distilled variants at most layers by cosine similarity, some evidence for attention alignment with the teacher. In **AG News**, however, the pattern is reversed: B is comparable to, and at several layers more similar to, the teacher than any distilled variant. Both datasets thus suggest attention-level similarity to the teacher is not a reliable signature of distillation, since the non-distilled baseline is nearly as close, or closer, to the teacher's attention geometry.

By contrast, Frobenius distance and JS divergence (Table V) are nearly indistinguishable across all four models at every layer on both datasets, following the same profile: near-perfect alignment at $i = 0$, peaking at $i = 2$–3 (Frobenius 34–41 out of a theoretical maximum of 64; JS 0.34–0.50), then stabilizing by $i = 4$–10. At an even finer grain, per-head attention-similarity analysis shows B numerically indistinguishable from the three distilled students on *both* datasets – the clearest evidence that per-head similarity to the teacher is insensitive to whether a model was distilled.

Table VI reports Participation Ratio (PR) and entropy-based effective rank (ER) on both datasets, at the same layers as Table V. On both measures, P_{avg} stays close to rank-one throughout, while the four students show no consistent ranking: on **AG News**, B trends from lowest to highest as depth increases on both PR and ER; on **IMDb**, the relative order among B, S_{bad}, S_1, and S_2 instead shifts from layer to layer. Spectral complexity is therefore not obviously ordered by distillation strength on either dataset.

Table III: **LIME** stability (mean pairwise Jaccard@10 across 5 seeds, $N = 20$), both datasets.

Dataset	P	B	S_{bad}	S_1	S_2
IMDb	0.548	0.587	0.684	0.611	0.710
AG News	0.774	0.669	0.639	0.615	0.688

Table IV: **LIME**, **SHAP**, and **Integrated Gradients** (IG, zero baseline) Jaccard@10, Spearman ρ, and Faithfulness@10 vs. teacher, both datasets ($N = 100$, $k = 10$).

Dataset	Model	LIME			SHAP			IG		
		Jac.@10	ρ	Faithf.	Jac.@10	ρ	Faithf.	Jac.@10	ρ	Faithf.
IMDb	P	–	–	0.172	–	–	0.118	–	–	0.010
IMDb	B	0.183	0.248	0.083	0.117	0.239	0.025	0.793	-0.005	0.010
IMDb	S_{bad}	0.170	0.216	0.202	0.123	0.238	0.093	0.778	0.028	0.085
IMDb	S_1	0.136	0.189	0.177	0.096	0.245	0.058	0.814	0.010	0.025
IMDb	S_2	0.212	0.299	0.267	0.176	0.333	0.120	0.787	0.039	0.019
AG News	P	–	–	0.138	–	–	0.003	–	–	0.054
AG News	B	0.336	0.321	0.306	0.315	0.270	-0.003	0.891	0.022	0.067
AG News	S_{bad}	0.348	0.280	0.402	0.317	0.275	0.000	0.883	0.040	0.163
AG News	S_1	0.342	0.307	0.299	0.320	0.267	0.000	0.876	-0.036	0.138
AG News	S_2	0.358	0.343	0.432	0.311	0.262	-0.001	0.888	0.013	0.340

Table V: Cosine similarity, Frobenius distance, and Jensen–Shannon divergence between each student's attention map and the teacher's P_{avg}, across layers i (Frobenius theoretical maximum $\sqrt{2n} = 64$ for $n = 2048$).

Dataset	i	Cosine				Frobenius				JS			
		B	S_{bad}	S_1	S_2	B	S_{bad}	S_1	S_2	B	S_{bad}	S_1	S_2
IMDb	0	0.933	0.923	0.937	0.923	1.265	1.356	1.236	1.353	0.011	0.018	0.011	0.019
IMDb	2	0.287	0.417	0.455	0.629	36.284	35.659	35.436	33.949	0.427	0.390	0.378	0.342
IMDb	3	0.228	0.433	0.502	0.687	39.117	38.205	37.842	36.461	0.493	0.438	0.433	0.386
IMDb	4	0.154	0.370	0.419	0.654	26.435	25.500	25.077	22.947	0.378	0.307	0.297	0.255
IMDb	7	0.133	0.260	0.479	0.381	18.040	17.391	16.082	16.808	0.369	0.270	0.236	0.251
IMDb	10	0.135	0.271	0.234	0.164	21.310	20.661	20.850	21.184	0.410	0.303	0.308	0.323
AG News	0	0.981	0.965	0.968	0.975	1.080	1.460	1.390	1.220	0.005	0.007	0.006	0.006
AG News	2	0.425	0.318	0.231	0.367	39.220	40.160	40.640	39.640	0.450	0.479	0.503	0.468
AG News	3	0.435	0.388	0.222	0.228	37.780	38.510	39.480	39.450	0.416	0.431	0.474	0.466
AG News	4	0.518	0.345	0.290	0.345	21.420	22.960	23.260	22.910	0.250	0.230	0.226	0.221
AG News	7	0.272	0.343	0.333	0.278	18.050	17.520	17.570	17.910	0.312	0.229	0.221	0.249
AG News	10	0.311	0.278	0.226	0.264	19.330	19.520	19.820	19.620	0.300	0.237	0.260	0.253

F. Discussion

Taken together, these results indicate that, at least in the studied settings, standard KD does not directly induce alignment of either explanatory mechanisms or internal representations between teacher and student. Both properties instead appear to emerge from the shared dataset and task rather than from the distillation term itself. Distillation's only clearly attributable effect is distributional: it reliably pulls the student's output distribution toward the teacher's (Table II). Accuracy itself is barely affected by distillation, and inconsistently so across datasets – but these remain well-behaved, competitive classifiers, worth studying on those terms alone.

This dissociation is not incidental: if explanatory or representational alignment were induced by the distillation term itself, both should track distillation strength (i.e. KL fidelity) directly. On **IMDb**, S_1 attains *lower* KL than S_2 (Table II) yet is markedly less **LIME/SHAP**-aligned with the teacher, the opposite of what a distillation-driven account would predict; **Integrated Gradients** shows no consistent best-aligned student at all, despite the same spread in KL fidelity. On **AG News**, B shows no disadvantage, and at several layers an advantage, in attention-level similarity to the teacher (Table V), despite having the highest accuracy of all five models. Neither similarity scales with distillation strength, reinforcing that the shared task and data are the more likely source of teacher-student similarity.

G. Does Distillation Add Value Beyond Depth Reduction?

Table VII summarizes the picture across both datasets. Distillation's only robust, task-independent effect is pulling the student's output distribution toward the teacher's. Its effects on accuracy, explanatory alignment, and fine-grained representational similarity are comparatively small, inconsistent across datasets, and, for attention geometry, largely absent once a non-distilled baseline is available for comparison.

H. Limitations

This study covers only two English text-classification datasets of modest label cardinality; whether the task-dependence between them (Table VII) generalizes to other modalities or higher-cardinality tasks is untested. The (α, β) grid is narrow: only two points beyond S_1/S_2 were tested ($\alpha = 1$ for B, $\alpha = 0.9$ for S_{bad}), and the opposite extreme (pure distillation, $\alpha = 0$) was never tested. Underlying all of this is a hardware constraint: every experiment ran on a single consumer GPU (RTX 5070 Ti, 16GB), bounding how many datasets, scales, and architectures could be explored.

IV. Conclusions

This work investigated whether functional similarity induced by knowledge distillation extends to internal representations and explanatory mechanisms, using a family of models independently on two tasks: binary **IMDb** sentiment classification and four-class **AG News** topic classification. In these comparatively simple settings, the answer is negative rather than definitively so: every student, including the baseline,

Table VI: Participation Ratio (PR) and entropy-based effective rank (ER), at layers i, both datasets.

Dataset	i	PR					ER				
		B	S_{bad}	S_1	S_2	P_{avg}	B	S_{bad}	S_1	S_2	P_{avg}
IMDb	0	2.6	2.6	2.5	2.7	3.3	60	117	96	107	309
IMDb	2	3.4	3.0	2.5	1.7	1.0	278	298	230	180	5
IMDb	3	2.9	3.3	2.5	2.0	1.0	328	398	304	361	7
IMDb	4	5.0	4.4	4.3	2.4	1.0	315	437	517	485	20
IMDb	7	13.7	13.4	7.5	7.1	1.0	505	555	505	488	84
IMDb	10	9.7	8.6	17.0	12.5	1.0	491	509	551	528	129
AG News	0	1.3	1.3	1.4	1.4	1.4	13	14	16	18	24
AG News	2	1.6	1.4	1.7	2.0	1.0	29	14	31	39	1
AG News	3	1.9	1.4	1.9	2.1	1.0	40	19	32	51	2
AG News	4	1.9	1.4	1.8	1.9	1.0	42	20	47	55	3
AG News	7	3.4	1.5	1.8	1.6	1.0	62	32	47	47	5
AG News	10	3.3	1.6	1.7	1.8	1.0	66	28	40	43	6

Table VII: Does distillation help beyond training a smaller model?

Property	**IMDb**	**AG News**
Accuracy	Mixed (S_2 only)	Hurts ($B >$ all)
KL fidelity	Robustly helps	Robustly helps
Explanation align.	Modest (S_2 best)	Modest (S_2 best)
Represent. align.	Weak ($B \approx$ others)	Absent (B competitive)

assigns importance to irrelevant tokens regardless of accuracy, and B is indistinguishable from the distilled students at the level of individual attention heads on both datasets; even at the level of full attention maps, distillation offers at best weak, inconsistent evidence of an effect on **IMDb** and none on **AG News**. The one property that behaves consistently, independently of accuracy, is distributional (KL) fidelity to the teacher.

A striking pattern underlies this: B, S_{bad}, S_1, and S_2 converge toward similar attention structure despite spanning α from 0.5 to 1. All four share the same architecture, dataset, and cross-entropy training, with only S_{bad}, S_1, and S_2 receiving a distillation term. Since B still converges to the same structure as the distilled students, that convergence cannot be attributed to distillation alone. A natural next step, left for future work, is a student trained with *pure* distillation ($\alpha = 0$, no cross-entropy term): if its representations diverge from the $B/S_{bad}/S_1/S_2$ cluster, that would implicate cross-entropy over the dataset, rather than distillation, as the source of this convergence.

Acknowledgment

The writing of this manuscript was partially assisted by generative language models used exclusively for stylistic revision and textual reformulation tasks: Claude Sonnet 4.6 [1] (Anthropic).

References

[1] Anthropic. Claude sonnet 4.6 [Large language model]. https://www.anthropic.com, 2025. Used as a generative AI tool for language editing and writing assistance.

[2] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4443–4458, 2020.

[3] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International journal of computer vision*, 129(6):1789–1819, 2021.

[4] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[5] Sarthak Jain and Byron C Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, 2019.

[6] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information theory*, 37(1):145–151, 1991.

[7] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.

[8] Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 142–150, Portland, Oregon, USA, June 2011. Association for Computational Linguistics.

[9] Stefano Recanatesi, Serena Bradde, Vijay Balasubramanian, Nicholas A Steinmetz, and Eric Shea-Brown. A scale-dependent measure of system dimensionality. *Patterns*, 3(8), 2022.

[10] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[11] Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *2007 15th European signal processing conference*, pages 606–610. IEEE, 2007.

[12] Timo Schick and Hinrich Schütze. Exploiting cloze questions for few shot text classification and natural language inference. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 255–269, 2021.

[13] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017.

[14] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[15] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 11–20, 2019.

[16] Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*, 2024.

[17] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28, 2015.

[18] Yang Zhou, Xu Gao, Zichong Chen, and Hui Huang. Attention distillation: A unified approach to visual characteristics transfer, 2025.